

Experiment No.4

Title: Calculation of correlation coefficient

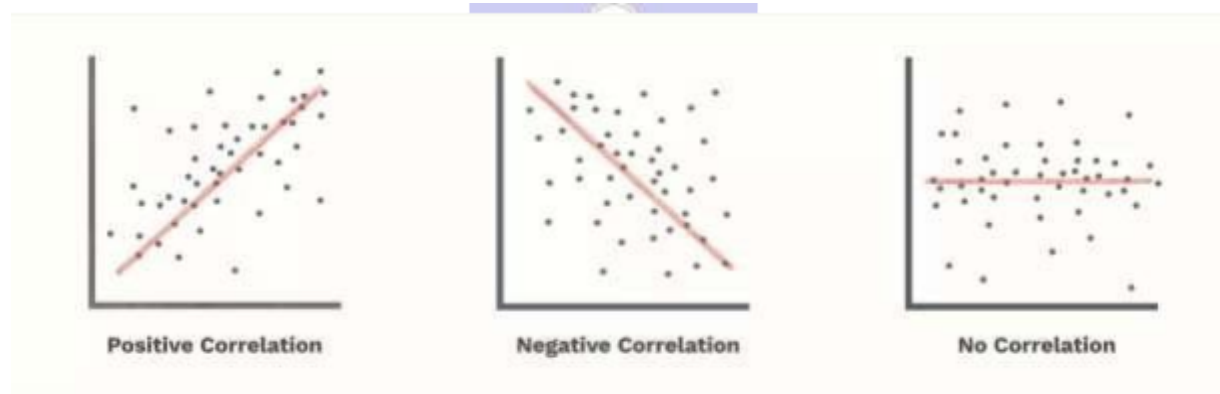
Aim: To write a program to compute correlation coefficient and plot the scatter plot

Resources needed: Python 3.x environment (Jupyter Notebook / Google Colab etc.) or R language environment

Theory:

Correlation coefficient:

The correlation coefficient is a statistical measure of the strength of a linear relationship between two variables. Its values can range from -1 to 1. A correlation coefficient of -1 describes a perfect negative, or inverse, correlation, with values in one series rising as those in the other decline, and vice versa. A coefficient of 1 shows a perfect positive correlation, or a direct relationship. A correlation coefficient of 0 means there is no linear relationship. These types are shown in figure below.



The sample correlation coefficient can be represented with a formula:

$$\begin{array}{c}
 \text{Sample Correlation Coefficient} \downarrow r = \frac{\sum [(x_i - \bar{x})(y_i - \bar{y})]}{\sqrt{\sum (x_i - \bar{x})^2 * \sum (y_i - \bar{y})^2}} \\
 \begin{array}{l}
 \text{Summation: "Take The Sum Of"} \downarrow \sum \\
 \text{Value of X} \downarrow x_i \\
 \text{Mean of X Variable} \downarrow \bar{x} \\
 \text{Value of Y} \downarrow y_i \\
 \text{Mean of Y Variable} \downarrow \bar{y} \\
 \text{Sum of the squared deviations for X} \downarrow \sum (x_i - \bar{x})^2 \\
 \text{Sum of the squared deviations for Y} \downarrow \sum (y_i - \bar{y})^2 \\
 \text{Square Root} \uparrow \sqrt{}
 \end{array}
 \end{array}$$

Correlation coefficients are used in science and finance to assess the degree of association

between two variables, factors, or data sets. For example, as high oil prices are favourable for crude producers, one might assume that the correlation between oil prices and forward returns

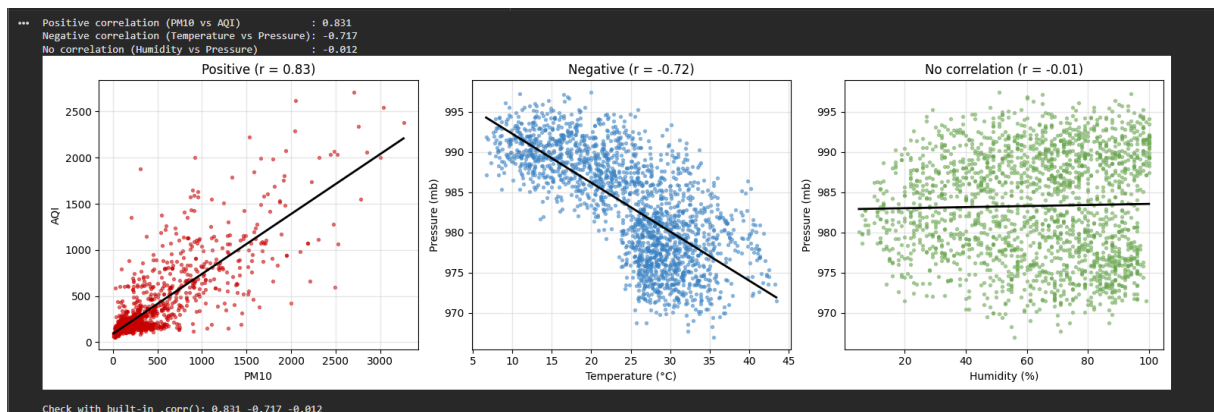
on oil stocks is strongly positive. Calculating the correlation coefficient for these variables based on market data reveals a moderate and inconsistent correlation over lengthy periods.

Procedure:

1. Step 1: Start the Program
2. Step 2: Import math package
3. Step 3: Define correlation coefficient function.
4. Step 4: Calculate correlation coefficient using formula for three types as positive, negative and no correlation. (without using in-built function)
5. Step 5: Print the result
6. Step 6: Plot the scatter plot.
7. Step 7: Stop the process

Results: (Program printout with output as per the format)

Ipynb file submitted on github



Questions:

1. What do the values of the correlation coefficient mean?

The correlation coefficient, or r , tells us how closely two things move together. It always sits between -1 and +1. If r is close to +1, the two things rise and fall together. In our data, PM10 and AQI have $r = 0.83$, which makes sense because when there's more dust in the air, the air quality gets worse. If r is close to -1, one goes up while the other comes down. Temperature and pressure showed this with $r = -0.72$, since hotter days tend to have lower air pressure. If r is close to 0, there's basically no connection. Humidity and pressure came out at -0.01, so knowing one tells you nothing about the other. The sign shows the direction, and the size shows how strong the link is. Roughly, 0.7 or more is strong, around 0.3 to 0.7 is moderate,

and below 0.3 is weak.

2. What are some limitations of correlation coefficient to consider?

The biggest catch is that correlation doesn't mean one thing causes the other. Two things can move together just because something else is affecting both. It also only picks up **straight-line** patterns, so if the data follows a curve, r can come out close to 0 even though there is a clear relationship. Outliers can throw it off too, because a few extreme values, like the very high AQI spikes in our data, can push r up or down. A single number can also hide what the data really looks like, which is why it's always a good idea to check the scatter plot as well. Finally, it only works for numbers, and a relationship that shows up in one place or time period might not hold in another.

Course Outcomes: (refer exp list)

Conclusion:

In this experiment, we calculated the correlation coefficient by writing our own function from the formula, without using any built-in function. We tried it on Delhi's 2025 weather and air-quality data. PM10 and AQI had a strong positive link ($r = 0.83$), temperature and pressure had a strong negative link ($r = -0.72$), and humidity and pressure had almost no link ($r = -0.01$). The scatter plots made these three patterns easy to see at a glance, and our results matched pandas' built-in function exactly. Overall, the experiment showed how correlation helps us spot relationships in data, and also reminded us that it can't prove that one thing causes another.

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of faculty in-charge with date

References:

Books/ Journals/ Websites:

1. S.C. Gupta, V. K. Kapoor, *Fundamentals of Mathematical Statistics*, Sultan Chand and Sons, 2020.
2. P. N. Tan, M. Steinbach, Vipin Kumar, *Introduction to Data Mining*, Pearson Education, 2016.
3. Han, Kamber, *Data Mining Concepts and Techniques*, Morgan Kaufmann, 3rd Edition, 2012